WISP-DC: A Benchmark-Trace-Based Feasibility Framework for Workload-Specific AI Infrastructure Planning

Raazia Imran Reshamwala, Syed Muhammad Sheraz, Maham Faisal

Abstract

Setting an AI-capable data center and high performance compute farm demands an early infrastructure decision: which available configuration can handle a defined workload without breaking performance and resource limits? The outcome hinges on the workload. Training is controlled by time-to-quality, whereas online inference must balance throughput, tail-latency, and accuracy conditions simultaneously. This paper details WISP-DC, an explainable feasibility framework that splits empirical performance proof from planning assumptions and ranks only configurations that pass every declared hard constraint. The framework was validated through two replicable, trace-driven scenarios. Scenario 1 used 30 raw MLPerf Training v4.0 ResNet-50 logs covering six H100 configurations from 4 to 64 accelerators. Depending on the deadline, the minimum-accelerator recommendation shifted across 4 to 8, 16, or 64 GPUs; no tested system met a two-minute deadline. Scenario 2 used four MLPerf Inference v5.0 Llama 2 70B Server results. Six predefined cases detailed that 4,000 and 8,000 output tokens/s were feasible with one and two GH200 accelerators, respectively; 20,000 tokens/s was infeasible under a two-accelerator ceiling but feasible with eight H100 accelerators; and 32,000 tokens/s exceeded every evaluated result. After excluding three price-dependent tests for which the underlying regional rate could not be separately logged, the preserved implementations passed 24 evidence-supported automated checks and executed 99 inference sensitivity cases. Power, rack, and energy calculations are reported only where their evidence boundary is met; financial optimization is not claimed. The study simply authenticates the framework's decision logic, rather than claiming new execution speeds or whole data center layout specifications.

Keywords: AI infrastructure, capacity planning, feasibility analysis, MLPerf, workload-aware systems, training, inference, explainable decision support.

1. Introduction

AARTIFICIAL-INTELLIGENCE platforms concentrate massive processing power within a compact physical footprint. This makes the initial scoping strategy crucial: a con-figuration that is insufficient fails to meet the service target, whereas one that is excessively ample squanders limited accelerators and may create avoidable power, space and cost overheads. The International Energy Agency forecasts global data-center electricity utilization to increase to roughly 945 TWh by 2030 in its base case, with accelerated servers accounting for a significant

fraction of the increase [1]. Earlier global analysis also showed why simple proportional extrapolation is flawed: demand can expand while enhancements in hardware, software, and utilization moderate energy growth [2]. In other words, infrastructure should be sized from the work it must deliver, not from a generic label such as "AI server."

Training and inference intensify this arbitration in dif-ferent paradigms. A training system must reach a precison target before a deadline. Adding accelerators can shorten the run, but the gain need not be linear and the larger system can impose a higher instantaneous power envelope. An online inference system instead faces a parallel set of requirements: request throughput, time to first token, time per output token, quality, and a allocation threshould. A high-throughput result is not useful if it breaches the latency or accuracy condition. Production traces likewise show that large-model development has heterogeneous job dynamics and non-trivial resource-utilization constraints [3].

Published evaluations provide a practical empirical basis for feasibility studies when the researchers do not own dozens of current accelerators. MLPerf Training re-ports time-to-quality under a regulated benchmark protocol [4]; MLPerf Inference assesses performance under defined deployment scenarios and quality rules [5]. Yet a benchmark table does not answer a user's provisioning question by itself. The table must be screened by the user's constraints, alterna-tives must be ranked under an explicit rule, and the outcome must explain why each candidate passed or failed.

This paper proposes and evaluates WISP-DC, a rule-based, workload-first feasibility framework. It does not purport to train ResNet-50 or execute Llama 2 70B locally. Instead, it treats official MLPerf records as empirical logs and evaluates the planning logic around those records. The present study explores three research questions:

- RQ1: Can a single architecture express the different hard constraints needed for AI training and online inference?
- RQ2: Does the implemented decision logic return the minimum-resource feasible evaluated configuration, and a justified infeasible result when no candidate passes?
- RQ3: Which conclusions remain valid when per-formance evidence, engineering proxies, prices, and scenario assumptions are kept separate?

The paper details four specific points. First, it maps out an verifable workload-to-configuration process that decouples hard-constraint filtering from scoring. Second, it incorporates separate training and inference decision models without extrapolating performance for unverified systems. Third, it gives a reproducible evaluation artifact comprising source records, scenario inputs, generated outputs, manual checks, sensitivity cases, and automated tests. Fourth, it shows how rack, energy, heat, and cost information can be integrated without presenting a surrogate as a physical measurement. The evaluated purview is AI training and online AI inference retrofitting, return on investment, cooling infrastructure, and broad HPC or enterprise-workload verification are omitted.

The rest of the paper examines related work, defines the framework and evidence boundary, describes the experiments, presents the results, and discusses validity andoperational impacts.

2. Related Work

2.1. Energy and Capacity Planning

Data-center energy research spans component models, server models, network models, and whole-facility models. Dayarathna et al. survey these levels and emphasize that model selection depends on the purpose of the analysis [6]. Fan et al. show that the sum of nameplate ratings is not nec-essarily an accurate representation of attainable aggregate demand, although rated values remain useful for conserva-tive electrical planning [7]. These distinctions are especially important for accelerator-dense systems: operational energy, expected peak demand, and installed electrical capacity are different quantities.

Recent measurement work reinforces that point. Latif et al. measure an eight-H100 node during ResNet and Llama 2 training and report workload-specific demand be-low the rated node limit [8]. Vercellino et al. provide high-resolution profiles for generative-AI training, fine-tuning, and inference to support facility planning [9]. Such measure-ments can calibrate preliminary estimates, but transferring a value between non-identical systems introduces uncertainty. WISP-DC therefore labels each numerical field as measured benchmark evidence, vendor evidence, derived output, or declared assumption.

2.2. AI Benchmarks and Workload Characterization

MLPerf Training uses an end-to-end time-to-quality objec-tive, which is more suitable for training feasibility than raw samples per second alone [4]. MLPerf Inference defines deployment scenarios, performance metrics, and accuracy requirements so that systems can be compared under a com-mon workload definition [5]. The official v4.0 Training and v5.0 Inference repositories retain system disclosures, result logs, and implementation records [10], [11]. This provenance makes the records suitable for a trace-replay study.

Standardization does not eliminate context. Performance remains tied to the model, dataset, precision, software stack, benchmark version, and submitted hardware. Nor does a published score determine which system should be pur-chased: the answer depends on the user's demand and resource limits. WISP-DC treats a benchmark result as an observed point, not as a universal scaling coefficient.

2.3. Resource Management and the Remaining Plan-ning Gap

Energy-aware allocation studies have largely focused on runtime management. Beloglazov et al. propose consol-idation heuristics that reduce energy while maintaining service con-straints [12]. GreenCloud provides packet-level simulation of servers, switches, links, and workloads [13]. Thermal-aware placement shows that workload scheduling and temperature cannot always be separated [14]. These studies are valuable once a data center and its resources exist.

The question studied here occurs earlier: given a specific AI workload and a finite catalogue of evidenced systems, which candidates are technically feasible, which candidate is preferred under a declared ranking rule, and why? Ta-ble 1 positions this contribution. The novelty is not a new benchmark or power model; it is the traceable combination of workload-specific feasibility rules, evidence boundaries, and reproducibility checks.

2.4. Sustainable Planning and Decision Transparency

Sustainable data-center planning is broader than lowering one power value. Shuja et al. organize energy-efficient data-center techniques across computing infrastructure, resource management, thermal behavior, and energy supply [15]. Carbon Explorer similarly demonstrates that operational choices, embodied impacts, power sources, and system life-time can lead to different design conclusions [16]. These studies are a useful warning against converting a single benchmark score into a broad sustainability claim. The present work therefore uses the term feasibility in a precise sense: satisfaction of declared performance and resource constraints within the available evidence. Energy and cost are attached as secondary decision attributes only when comparable inputs exist.

Explainability is equally important because an infra-structure recommendation can fail for several unrelated reasons. A configuration may lack sufficient throughput, exceed the allowed accelerator count, miss a latency target, fail the required accuracy threshold, or violate an applicable rack or budget limit. A single composite score can conceal such a failure. WISP-DC instead keeps a Boolean pass/fail record for every hard constraint and exposes the ranking rule used after filtering. This design favors auditability over an opaque weighted score.

2.5. Evidence Reuse as an Evaluation Method

Trace-driven evaluation is appropriate when a research con-tribution concerns a decision process rather than a new hardware or learning algorithm. It allows the same evi-dence to be replayed under different requirements, makes boundary cases repeatable, and separates errors in source extraction from errors in recommendation logic. Its principal weakness is dependence on the external experiment: the researcher can verify the mapping applied to the trace but cannot purport to have independently reproduced the physical measurement. This paper incorporates that limita-tion unambiguously. The MLPerf logs are empirical inputs, and every claim about WISP-DC is confined to what the implemented transformations and tests establish.

3. Framework and Method

3.1. Scope and Evidence Boundary

WISP-DC acts as a simple checking step between what a user needs and a list of eligible systems. It answers a very specific question: does a tested setup meet every strict rule, and if a few do, which one does the standard tie-breaker choose? The tool never makes up an unseen cluster on the fly, guesses performance using flat scaling lines, or takes the place of official electrical, weight, and cooling blueprints "WISP-DC acts as a simple checking step between what a user needs and a list of eligible systems. It answers a very specific question: does a tested setup meet every strict rule, and if a few do, which one does the standard tie-breaker choose? The tool never makes up an unseen cluster on the fly, guesses performance using flat scaling lines, or takes the place of official electrical, weight, and cooling blueprints

The evidence ledger uses four classes:

Table 1: Positioning of WISP-DC Relative to Representative Work

Work	**Primary purpose**	**Workload evi-dence**	**Planning decision**	**Validation form**
MLPerf Training [4]	Standardized train-ing comparison	Time-to-quality	No user-specific selec-tion	Controlled bench-mark
MLPerf Inference [5]	Standardized infer-ence comparison	Throughput, latency, accuracy	No user-specific selec-tion	Controlled bench-mark
Beloglazov *et al.* [12]	Runtime consolida-tion	Simulated cloud de-mand	VM allocation	Simulation
GreenCloud [13]	Energy-aware net-work simulation	Configurable work-loads	Simulated alloca-tion/topology	Packet-level simula-tion
Latif *et al.* [8]	AI-node power char-acterization	Measured H100 node	No infrastructure rec-ommendation	Physical measure-ment
Proposed WISP-DC	Early feasibility deci-sion	Published MLPerf traces	Recommend an evi-denced candidate or return infeasible	Source reconciliation, cases, tests, sensitivi-ty

1. empirical benchmark evidence: MLPerf timestamps, throughput, latency, accuracy, and submitted sys-tem descriptions;
2. vendor evidence: form factor, weight, device memory, and conservative electrical specifications;
3. derived outputs: differences, trimmed means, feasi-bility flags, energy arithmetic, and heat conversion; and
4. scenario assumptions: deadlines, accelerator ceilings, PUE, rack limits, and run frequency.

Empty data simply stays empty. For instance, the four-GPU training choice is still kept for the speed sorting step even though a matching server power number was not there; the code simply skips running power calculations for it. The hardware cross-checks follow the same rule. NVIDIA's H100 specification is used as a device-rating refer-ence [17], while the Dell XE9680 rack-integration guide and HPE XD670 QuickSpecs support model-specific physical and conservative electrical fields [18], [19]. These documents do not substitute for workload performance or whole-system energy measurements.

3.2. Architecture

Figure 1 presents the evaluated decision path. The user input becomes a scenario-specific workload profile. The training or inference model then filters empirical candidates. A feasible set is ranked; an empty set produces an explicit infeasible outcome. Supporting rack, power-proxy, and cost checks are added only when their data boundary is satisfied. The prototype documentation also contains broader ta-bles for servers, GPUs, CPUs, storage, networking, racks, PDUs, UPS equipment, cooling-related hardware, and power distribution. These tables are valuable for implement-ing a future catalogue service, but they were not used as performance evidence in S1 or S2. Several are heterogeneous compilations rather

than analysis-ready relational tables, two uploaded power-distribution files are duplicates, and some fields are estimates or "not stated." In the present paper, only matching OEM server/GPU facts support static cross-checks. The broader catalogue is therefore shown as a separate prototype layer rather than merged with the experimental dataset.

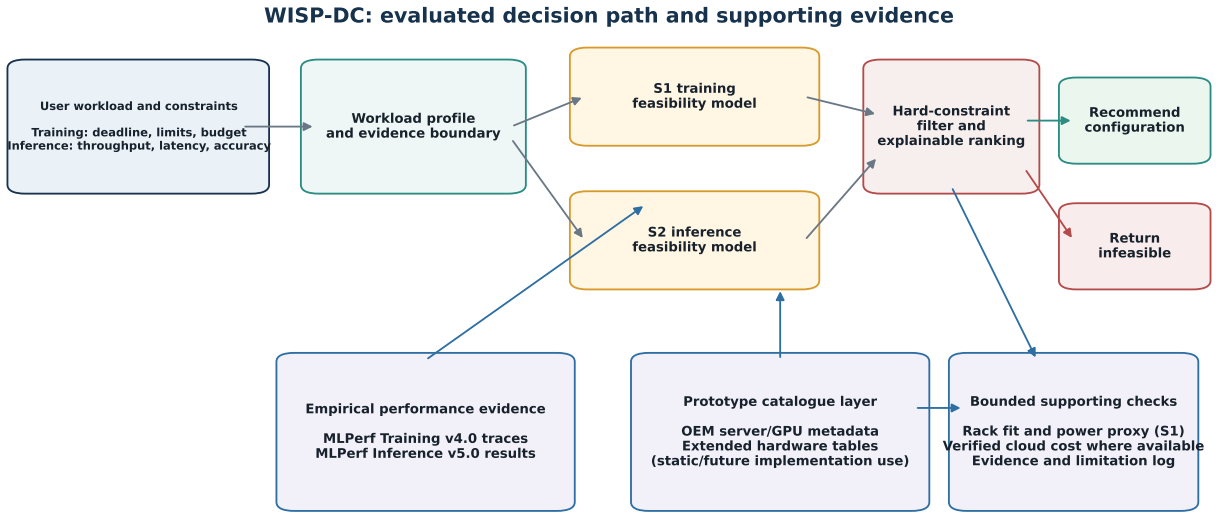


Figure 1: WISP-DC architecture. Solid decision stages were evaluated through S1 and S2. The broader prototype catalogue is documented separately and is not presented as benchmark validation.

The evaluated path in Fig. 1 sits within a broader archi-tecture proposed for the full prototype. Figure 2 shows how workload profiling, infrastructure sizing, energy, cost, risk, existing-facility assessment, evidence sources, and reporting are intended to interact. It is included as a conceptual system map rather than an implementation claim. The present experiments exercise only the S1 and S2 feasibility path described in this paper; the existing-facility workflow, com-plete cost and cooling engines, external interfaces, and ex-tended reporting functions remain future prototype scope.

3.3. Decision Procedure and Explanation Record

Both routines use an identical control flow. Whilst the actual calculations are very different, you can see how the process works from Table 2. We've written this without any form of language so you don't get confused with any language semantics. Firstly, there is a screening process. Any hardware configurations which cross any strict boundaries will be excluded from being used. Every check will write a line of information to the audit log detailing what the empirical value was, what the target value was, whether it passed or failed and where the data originated from. This makes checking our sizing decisions easy to do. If there is a complete failure on validation, we still gain some debugging information. We have written the software to avoid any of these common shortcuts. Firstly, the code doesn't take

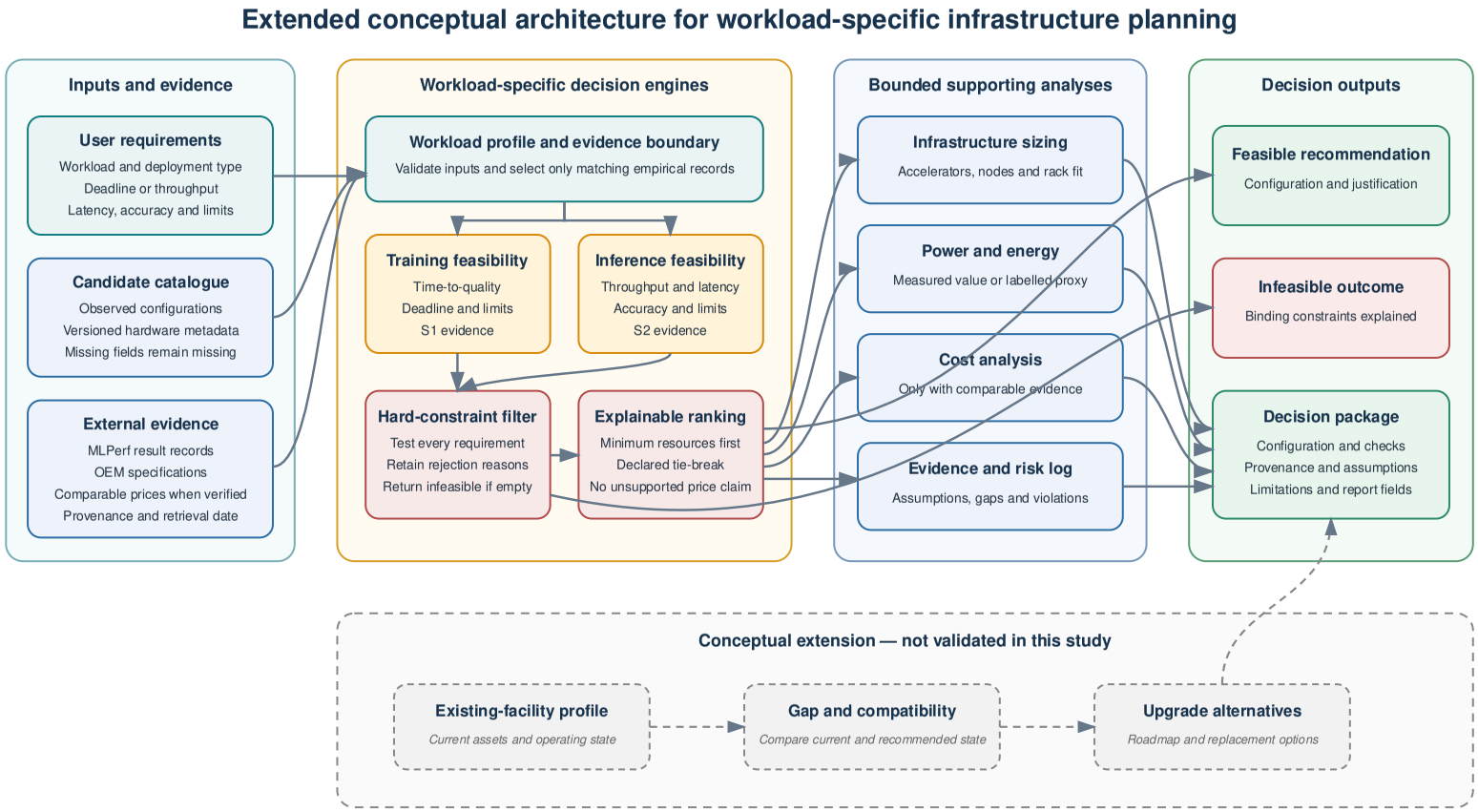


Figure 2: Extended conceptual architecture for the wider infrastructure-planning prototype. The diagram records the intended end-to-end design while distinguishing future extensions with dashed borders and connectors. This paper validates only the workload-specific S1/S2 feasibility path and its bounded supporting checks; the existing-facility, complete cost and cooling, external-integration, and extended-reporting functions are not presented as implemented or experimentally validated.

a runtime on a single GPU and extrapolate to work out scaling metrics for a large cluster. Secondly, if there isn't data centre cost sheets or power information this isn't just assumed to be zero. Lastly, we do not allow soft scoring to hide any failures on the infrastructure. All of these can be seen in the automated testing repo.

Table 2: Common Decision Procedure

Step	Operation
1	Validate units, positive limits, and required fields.
2	Select the workload-specific empirical catalogue.
3	Evaluate every applicable hard constraint for every candidate.
4	Store passing candidates in the feasible set and retain rejection reasons for all others.
5	If the feasible set is empty, return *infeasible* with the binding constraints.
6	Otherwise apply the declared lexicographic ranking rule.
7	Attach only those rack, power, energy, or cost outputs whose evidence requirements are satisfied.
8	Export the decision, provenance, assumptions, and limitation flags.

3.4. Training Feasibility Model

Let $\mathcal{C}_T$ be the set of observed training configurations. Candi-date c has an observed time-to-quality t_c, accelerator count g_c, and optional engineering fields. For deadline D, and any declared resource, rack, or budget limits collected in $\boldsymbol{b}$, the feasible set is

$$\mathcal{F}_T(D, \boldsymbol{b}) = \{c \in \mathcal{C}_T : t_c \leq D \ \wedge \ h_c(\boldsymbol{b}) = 1\}, \tag{1}$$

where $h_c = 1$ means that every applicable non-performance constraint passes. If $\mathcal{F}_T$ is non-empty, the default recom-mendation is

$$c_T^* = \arg\min_{c \in \mathcal{F}_T} (g_c, t_c), \tag{2}$$

using lexicographic order: fewest accelerators first, then shorter observed time. This is a minimum-resource rule, not proof of minimum purchase price.

For each submitted configuration, duration is calculated from the official start and stop timestamps. Following the benchmark aggregation represented in the source work-book, the largest and smallest of five durations are removed and the remaining three are averaged:

$$\bar{t}_c = \frac{1}{3} \sum_{i \in \mathcal{I}_c} t_{c,i}, \qquad \mathcal{I}_c = \{1, \ldots, 5\} \setminus \{\arg\min t, \arg\max t\}. \tag{3}$$

The raw-run audit remains independent of this arithmetic. It detected one log marked aborted in the Dell eight-GPU set; the record is retained and disclosed rather than silently deleted. Because it was also the maximum duration, it did not enter the three-run mean.

3.5. Inference Feasibility Model

Let candidate $c \in \mathcal{C}_I$ have measured output-token through-put q_c, accelerator count a_c, p99 time to first token f_c, p99 time per output token p_c, and quality vector $\boldsymbol{r}_c$. For required throughput Q, maximum accelerators A, latency limits F and P , and quality thresholds $\boldsymbol{R}$,

$$\mathcal{F}_I = \{c \in \mathcal{C}_I : q_c \geq Q,\ a_c \leq A,\ f_c \leq F, \\ p_c \leq P,\ \boldsymbol{r}_c \succeq \boldsymbol{R},\ \text{result}_c = \text{VALID}\}. \quad (4)$$

The recommendation is

$$c_I^* = \arg\min_{c \in \mathcal{F}_I} (a_c, q_c - Q). \quad (5)$$

Thus the framework first minimizes accelerator count and then unused measured through-put. This deterministic tie-break explains why the eight-H100 system is preferred over the eight-H200 system at 20,000 and 30,000 output tokens/s. It does not assert that H100 is cheaper.

3.6. Bounded Engineering Calculations

S1 uses a rough operating-power estimate based on sourced records of 6.26 kW for a comparable eight-H100 ResNet training node [8]. For a valid setup with proxy power P_c (kW) and runtime t_c (h),

$$E_{\text{IT},c} = P_c t_c, \quad (6)$$

$$E_{\text{facility},c} = \text{PUE}\, E_{\text{IT},c}, \quad (7)$$

$$H_c = 3412.14 P_c, \quad (8)$$

where energy is in kWh and H_c is the computed heat rate in BTU/h. PUE was kept at 1.4 for the starting point and altered from1.2 and 1.6. The proxy was multiplied by node count for the two-node HPE and eight-node OCI setups. This is simple math for sensitivity analysis, not a direct telemetry of those submitted systems. Rack safety demands vendor maximum or inferred redundant protection instead of the operating-energy proxy. The distinction stops typical run power from being mistaken for an electrical design limit.

If a future deployment provides a separately verifiable and comparable accelerator-hour rate, the corresponding run cost can be computed as

$$C_c = g_c\, t_c\, \rho, \quad (9)$$

where ρ is the correct accelerator-hour rate. The equation maps out a future cost extension; it is not evaluated numerically in this study because a dated, configuration-matched regional rate was not independently archived. On-premise capital-cost ranking is also omitted as comparable quotations were missing and hard to find.

3.7. Reproducibility Design

Each workflow has independent Python code base, machine-readable benchmark parameters, stated test cases, generated CSV or JSON outputs, and manual checks. The program skips assuming an untested configuration. Each disqualification records the failing condition. Validation combines: (i) source reconciliation; (ii) independent arithmetic for selected outputs; (iii) expected-result cases; (iv) deliberate invalid-input and hard-constraint tests; (v) workbook-to-Python comparison; and (vi) sensitivity sweeps. This method finds the transformation from evi-dence to decision, which is the real academic takeaway.

4. Experimental Design

4.1. Data Acquisition, Extraction, and Reconciliation

The measured figures were not pulled from a summary chart alone. For S1, the spec sheets and each raw result file were saved alongside their original MLCommons source links. Start time, stop time, final accuracy, global batch size, and run status were parsed into a tracked JSON sheet. The calculated runtime was recalculated from the timestamps Instead of typing it by hand. For S2, the performance summary, accuracy output, and system JSON for each chosen submission were Kept as distinct links in the benchmark CSV. The simulator processes these local, reviewable extracts; never relies on a live web request while running.

Three reconciliation layers cut down Data entry errors. First, the raw values are compared with the official result records. Second, a formula-based workbook separately tracks the same cases. Third, the Python output is compared with the expected workbook decision. The source record, workbook, and program therefore have distinct roles: source evidence provides the observations, the work-book makes the calculation visible to a reviewer, and Python establishes repeatable execution.

4.2. S1: ResNet-50 Training

S1 relies on MLPerf Training v4.0 closed-division ResNet-50 v1.5 results on ImageNet with the 75.90% quality target [10]. Six H100 setups were chosen: Dell systems with four and eight GPUs, HPE Cray XD670 systems with eight and sixteen GPUs, and Oracle bare-metal configurations with eight and sixty-four GPUs. Five direct output files per configuration yielded 30 duration observations. Table 4 lists the derived representative times.

The core sensitivity parameter is the completion cutoff: 3, 5, 8, 10, 15, 20, 30, and 60 minutes. Additional con-trolled validations evaluate a two-minute infeasible deadline, rack-power rejection, PUE monotonicity, and explicit handling of cloud physical placement. No local ResNet-50 training was executed; acquiring the largest evaluated system would mandate 64 H100 accelerators.

The raw-run review also records dispersion before the trimmed aggregation. Table 5 details the coefficient of vari-ation (CV) across the five retrieved durations. Five setuos have CV below 0.16%, proving very consistent iterated times. Dell-8G is the exception at 2.64% because its fifth record is labeled aborted and has a duration of 13.944 minutes, compared with approximately 13.15–13.17 minutes for the four successful records. Disclosing this dif-ference matters: the final representative time is consistent after the predefined trimming rule, but the raw evidence is not described as five equivalent successful runs.

Table 3: Evidence Used in the Two Experimental Scenarios

Layer	**S1 training**	**S2 inference**	**Permitted claim**	**Claim not permitted**
Performance	30 MLPerf v4.0 raw training logs	Four MLPerf v5.0 Server results	Feasibility among the observed configurations	Locally measured performance or untested scaling
Quality	Final ResNet-50 accuracy and run status	VALID flag and ROUGE-1/2/L thresholds	Benchmark quality condition passed	General model-quality superiority
System	Six MLPerf system disclosures	Four MLPerf system disclosures	Accelerator/node identity of each submission	Interchangeability of different software stacks
Physical	Matching Dell/HPE/OEM documents where available	Static accelerator/server metadata only	Bounded form-factor or conservative capacity check	Complete facility, cooling, or electrical design
Power	Comparable measured eight-H100 ResNet node	No matched operational record retained	S1 scenario proxy with explicit transfer limitation	Direct power comparison across submitted systems
Price	No independently archived comparable rate	No comparable price set	General cost-extension equation only	Any cheapest-system or numerical run-cost claim

Table 4: S1 Evaluated Training Configurations

ID	**GPUs**	**Nodes**	**Time (min)**
Dell-4G	4	1	26.882
Dell-8G	8	1	13.163
HPE-8G	8	1	13.619
HPE-16G	16	2	7.930
OCI-8G	8	1	13.273
OCI-64G	64	8	2.483

Table 5: S1 Raw-Run Status and Dispersion

ID	**Successful**	**Aborted**	**CV (%)**
Dell-4G	5	0	0.052
Dell-8G	4	1	2.637
HPE-8G	5	0	0.085
HPE-16G	5	0	0.158
OCI-8G	5	0	0.050
OCI-64G	5	0	0.043

4.3. S2: Llama 2 70B Online Inference

S2 relies on four HPE closed-division MLPerf Inference v5.0 results for Llama 2 70B, the OpenOrca dataset, Server scenario, and the 99%-of-FP16 accuracy target [11]. Table 6 evaluates the fields imported into the simulation. Every row was marked VALID and passed the published TTFT, TPOT, and ROUGE thresholds. The study uses the p99 latency limits recorded with these results: 2.0 s for TTFT and 0.2 s for TPOT.

Six prescribed workload profiles vary throughput and maximum accelerator count. The throughput contraints are 4,000, 8,000, 20,000, 20,000, 30,000, and 32,000 output tokens/s; the corresponding accelerator ceilings are 1, 2, 2, 8, 8, and 8. A 99-case matrix then varies throughput and resource limits to evaluate decision boundaries. S2 does not calculate power or financial cost because equivalent operational power and comparable purchase-price evidence were missing.

Table 6: S2 Official MLPerf Inference v5.0 Inputs

Configuration	Accelerators	Output tok/s	p99 TTFT (s)	p99 TPOT (s)	ROUGE-1	Result
HPE-GH200-1	1	4,327.68	1.8895	0.1909	44.5531	VALID
HPE-GH200-2	2	8,674.58	0.8064	0.1712	44.5503	VALID
HPE-H100-8	8	30,498.87	1.2950	0.1429	44.4303	VALID
HPE-H200-8	8	31,281.09	0.6660	0.1685	44.5663	VALID

5. Results

5.1. S1 Training Feasibility

Figure 4 depicts the profiled training times. Increasing scale reduced time-to-quality threshold, but the allocation was dictated by the latency bound rather than by speed alone. The 30- and 60-minute scenarios nominated Dell-4G. The 15- and 20-minute runs selected Dell-8G. The 8- and 10-minute instances mandated HPE-16G, and the 3- and 5-minute cases mandated OCI-64G. A two-minute deadline proved unattainable. These shifts answer RQ1 and RQ2 for training workloads: the same workload binds to different resources when the service requirement changes. Figure 3 renders these decision limits visible. A dead-line underneath an observed execution time alters the feasible set discontinuously; the selection is there-fore a step function rather than a smooth hardware-sizing curve. This behavior is intentional. Between observed coordinates, the framework does not fabricate a fractional accelerator requirement or estimate a new system. Table 7 details the complete feasibility matrix grid. A verification indicates that the tracked time duration is within the contraint; the selected row is the feasible configuration with the fewest GPUs and then the shorter time.

Within the eight-GPU cluster, Dell-8G, HPE-8G, and OCI-8G finished within 13.163, 13.619, and 13.273 minutes, respec-tively. These are indicative cross-system observations, not controlled vendor rankings, because the submitted stacks differ. Within-family comparisons are more interpretable: moving HPE from 8 to 16 GPUs minimized time from 13.619 to

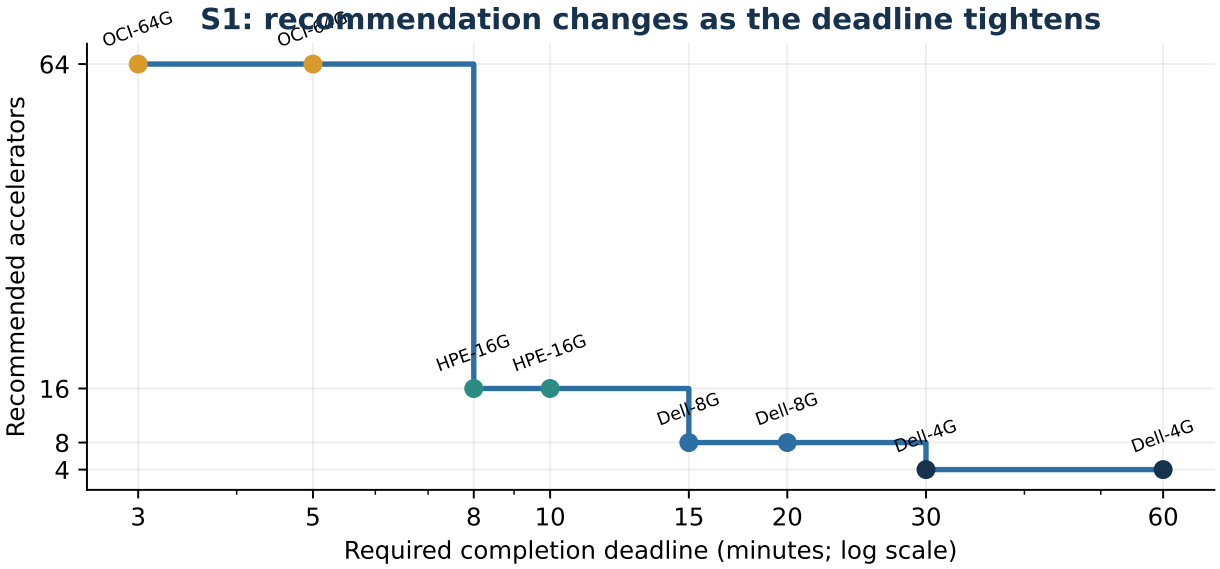


Figure 3: S1 deadline sensitivity. The recommendation changes only when another observed configuration becomes necessary.

Table 7: S1 Deadline Feasibility Matrix

Deadline	Dell-4G	Dell-8G	HPE-8G	HPE-16G	OCI-8G	OCI-64G	Min. GPUs	Recommendation
3 min	–	–	–	–	–	Pass	64	OCI-64G
5 min	–	–	–	–	–	Pass	64	OCI-64G
8 min	–	–	–	Pass	–	Pass	16	HPE-16G
10 min	–	–	–	Pass	–	Pass	16	HPE-16G
15 min	–	Pass	Pass	Pass	Pass	Pass	8	Dell-8G
20 min	–	Pass	Pass	Pass	Pass	Pass	8	Dell-8G
30 min	Pass	Pass	Pass	Pass	Pass	Pass	4	Dell-4G
60 min	Pass	Pass	Pass	Pass	Pass	Pass	4	Dell-4G

7.930 minutes (a 41.8% lowered, rather than an optimal 50%), while moving OCI from 8 to 64 GPUs reduced time from 13.273 to 2.483 minutes (an 81.3% reduction, rather than the 87.5% optimal under strict eightfold scaling). The findings warns against using accelerator count as a linear performance predictor.

The base power surrogate generates facility-energy estimates of 1.923 kWh/run for Dell-8G, 1.989 for HPE-8G, 2.317 for HPE-16G, 1.939 for OCI-8G, and 2.902 for OCI-64G at PUE 1.4. Dell-4G is purposefully excluded since no equivalent four-GPU node power value was archived. The fastest option therefore did not have the lowest proxy energy per com-pleted run. This is a qualified comparison under a common cross-study calibration, not measured facility energy.

Table 8 illustrates the PUE sensitivity visibly. Shifting from PUE 1.2 to 1.6 increases every facility calculation by one third because workload execution window and IT surrogate power remain persistent. This is a valid internal coherance test, but it is not evidence

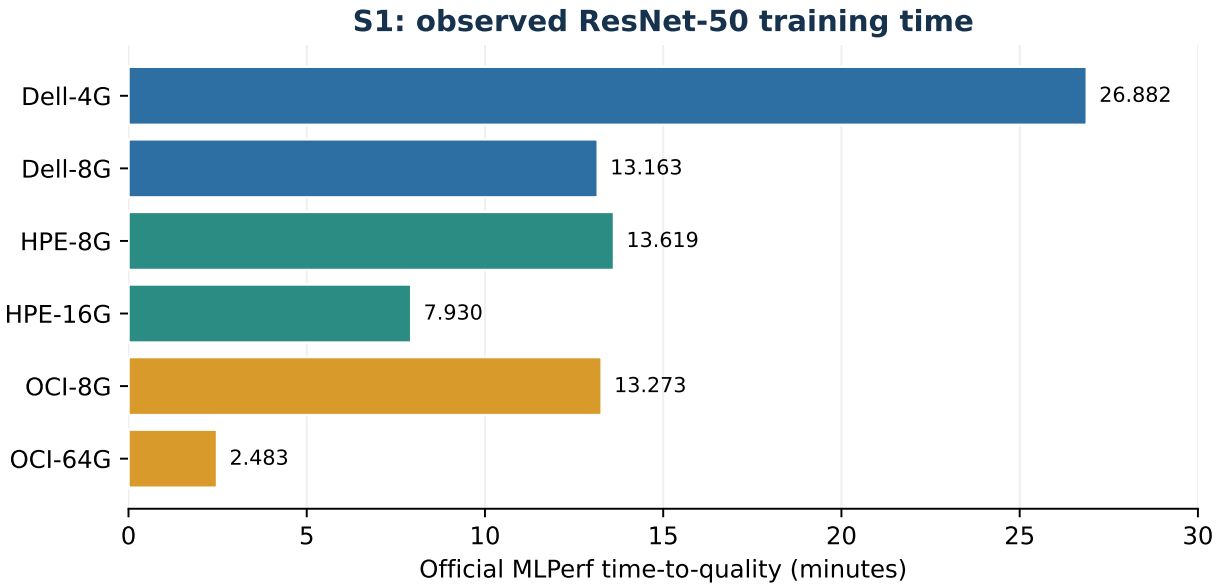


Figure 4: S1 representative time-to-quality derived from official MLPerf Training v4.0 raw timestamps. Lower is better.

that any profiled site actually performs at one of these PUE values. Annual figures in the workbook spreadhseet assume one equivalent completed run per day; the paper highlights per-run values because the annual frequency is a case-study assumption rather than an recorded operating timeline.

Table 8: S1 Facility-Energy Proxy Under PUE Sensitivity (kWh/run)

ID	PUE 1.2	PUE 1.4	PUE 1.6
Dell-8G	1.648	1.923	2.197
HPE-8G	1.705	1.989	2.274
HPE-16G	1.986	2.317	2.648
OCI-8G	1.662	1.939	2.216
OCI-64G	2.487	2.902	3.316

The rack sensitivity employs a different validation class. Dell-8G uses an 11.5-kW provisioning cap; HPE-8G and HPE-16G use 12 and 24 kW calculated net capacities. All three populate one assumed rack at 40, 80, and 120 kW bounds when space and weight also qualify. A knowingly 10-kW test drops the Dell node. The outcome showcases constraint behavior, not vendor validation of a particular rack deployment.

No quantitative cloud-cost comparison is documented because the exact regional price, service identifier, currency, and retrieval log could not be independently logged to the same evidence standard as the benchmark results. The framework hence reports technical feasibility and minimum-resource ranking only. It does not conclude which cloud or local system has the lowest monetary cost.

5.2. S2 Inference Feasibility

Figure 5 illustrates the four profiled throughput coordinates and the 20,000-token/s example threshold. Table 9 details the specified results. The one and two-GH200 configurations matched the 4,000- and 8,000-token/s scenarios. Neither could satisfy 20,000 tokens/s under a two-accelerator bound, so T3 accurately returned unattaunable. Modifying the ceiling to eight apermitted both H100-8 and H200-8 to pass. Equaion (5) nominated H100-8 because both use eight accelerators and H100-8 has the smaller surplus above the bound. The same rule selected H100-8 at 30,000 tokens/s. At 32,000 tokens/s, even H200-8 shortfalled by 718.91 tokens/s, producing another justified infeasible outcome.

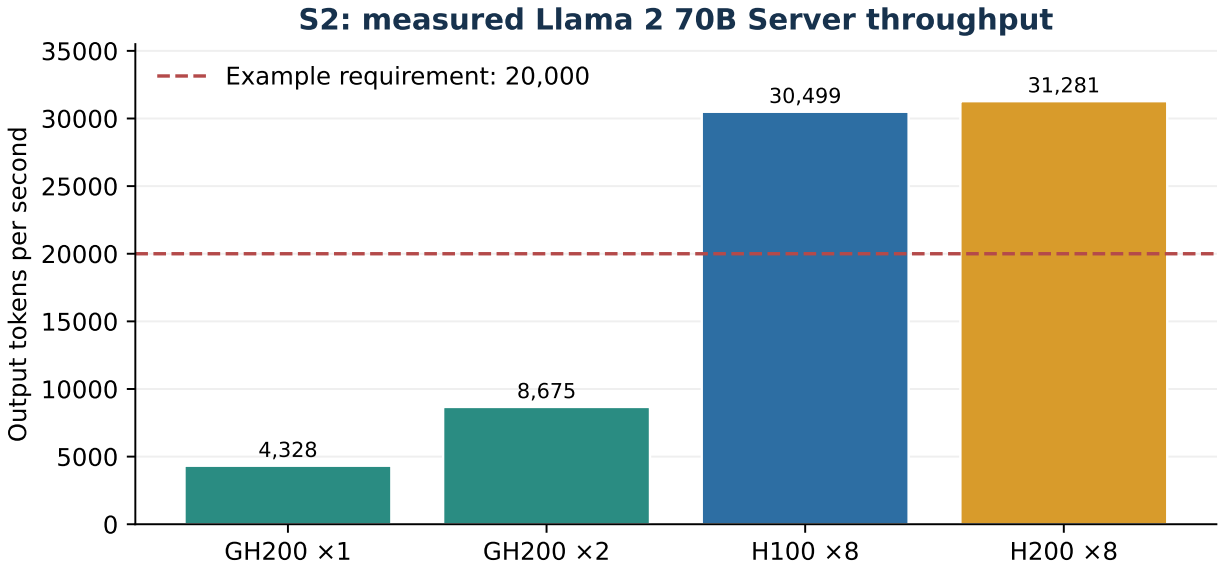


Figure 5: S2 measured output-token throughput. The dashed line illustrates T4's 20,000-token/s requirement; latency and accuracy must also pass.

Throughput exclusively did not determine feasibility. All four baseline records already met the 2.0-s TTFT bound, 0.2-s TPOT limit, validity flag, and quality thresholds. Separate tests restricted each condition until no candidate passed, verifying that the implementation treats them as hard constraints. The result resolves RQ1: the shared architecture can support inference, but only through a requirement vector different from training's deadline rule.

The 99-case sensitivity grid gives a vast view than the six isolated examples. Figure 6 varies the target rate from 2,000 to 34,000 output tokens/s and the accelerator ceiling across one, two, and eight devices. With single accelerator, feasibility terminates after the 4,000-token/s point because GH200-1 registers 4,327.68. With two accelerators, GH200-2 remains available through 8,000 tokens/s. Under the eight-accelerator ceiling, H100-8 becomes the preferred observed system from 9,000 through 30,000 tokens/s. At 31,000 to-kens/s, H100-8 is insufficient and H200-8 is selected. From 32,000 tokens/s onward, the observed catalogue is infeasi-ble. These transitions follow the measurements exactly; they are not rounded capacity forecasts.

Table 9: S2 Predefined Cases and Python Decisions

Case	Tok/s	Max acc.	Decision
T1	4,000	1	GH200-1
T2	8,000	2	GH200-2
T3	20,000	2	Infeasible
T4	20,000	8	H100-8
T5	30,000	8	H100-8
T6	32,000	8	Infeasible

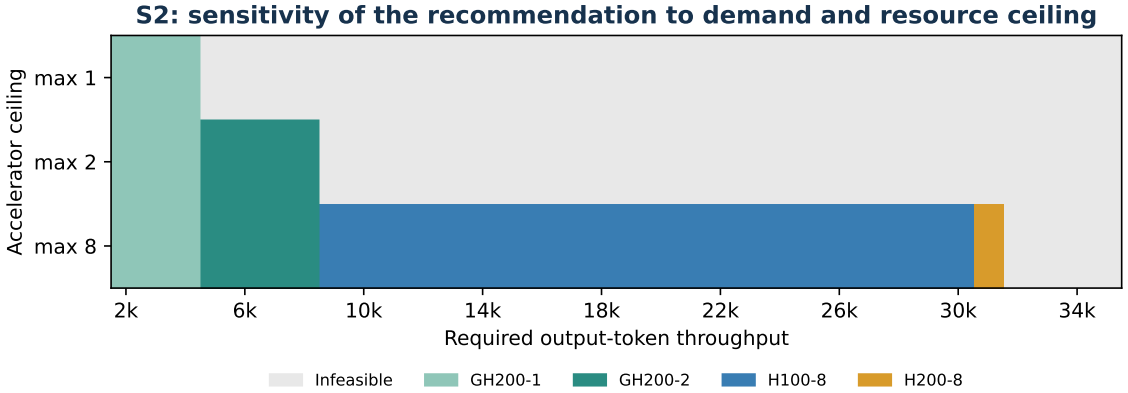


Figure 6: S2 sensitivity across 99 combinations of throughput requirement and accelerator ceiling. Grey cells are infeasible among the four observed candidates.

Table 10 itemizes the explanation record for T4. GH200-1 and GH200-2 fail throughput even though they clear the accelerator ceiling and baseline quality/latency parameters. Both eight-accelerator systems pass. H100-8 is isolated only after filtering, through the minimum-surplus tie-break. This logging distinguishes a disqualification basis from a ranking factor.

Table 10: Illustrative Candidate-Level Explanation for T4 (20,000 Output Tokens/s, Maximum Eight Accelerators)

Candidate	Throughput	Acc. limit	TTFT	TPOT	Quality	Outcome and reason
GH200-1	Fail	Pass	Pass	Pass	Pass	Rejected: observed throughput below 20,000
GH200-2	Fail	Pass	Pass	Pass	Pass	Rejected: observed throughput below 20,000
H100-8	Pass	Pass	Pass	Pass	Pass	Feasible and selected: eight accelerators, smaller surplus
H200-8	Pass	Pass	Pass	Pass	Pass	Feasible but not selected: equal count, larger surplus

5.3. Validation Summary

Table 11 condenses the replicability proof. Six data backed S1 tests were retained after the three cost sensitive cases were omitted. S2 cleared eighteen unit tests, replicated all six workbook cases, and produced 99 sensitivity records. Manual audits reproduced the Dell-8G energy arithmetic and extracted S2 throughput-per-accelerator and latency-buffer values. Invalid non-positive parameters were dropped. Tests also verified that cloud rack placement remains "provider managed," missing prices are not assumed, and S2 does not label its recommendationlowest cost.

The evaluation proves exact agreement be-tween source records, formulas, Python, and workbook cases. It does not separately verify the physical metrics disclosed by MLCommons. That distinction addresses RQ3: the evidence supports the framework's transformation and boundary handling, while the benchmark organization remains the source of the underlying performance measurements.

The checks span exceed predicted recommendations. S1 evaluates an impossible deadline, a single-candidate boundary, a rack-power rejection, visibility of an aborted log, mono-tonic PUE behavior, and cloud rack semantics. S2 checks record counts and exact throughput reconciliation; all six expected cases; the effect of relaxing an accelerator ceiling; deterministic tie handling; the narrow interval in which H200-8 is selected; requirements above the observed maximum; independent TTFT, TPOT, and accuracy rejections; manual arithmetic; the explicit absence of price optimization; and invalid non-positive input handling. The entire Catalogue is detailed in Appendix A.

Sensitivity and unit testing resolve distinct questions. The unit cases ask whether a defined threshold triggers the expected output. The sensitivity grid asks whether decisions remain coherent over a wider input space. Nei-ther dynamic demonstrates completeness of the candidate catalogue, but combined they mitigate the six headline cases merely reflect manual interpretation.

Table 11: Reproducibility and Validation Evidence

Check	**S1**	**S2**
Official evaluated configurations	6	4
Raw performance logs/records	30	4
Predefined result cases	8 deadlines	6 cases
Automated checks retained and passed	6/6	18/18
Workbook cases reproduced	Yes	6/6
Sensitivity cases	PUE/rack	99
Local model execution	No	No

6. Discussion

6.1. What the Two Scenarios Establish

The unified outcome is more pragmatic than a single "optimal GPU" statement. In S1, the core performance variable is elapsed time to a defined quality target. Conversely,

inside S2, the core metrics are output-token rate, tail latency, and accuracy under a Server workload. The framework architecture can be unifoed, but the feasibility predicate cannot. A generic accelerator-count formula would obscure this difference and invite unsubstantiated extrapolation.

The experiments further demonstrate why infeasibility is a valid planning outcome. T3 does not fail because the accelerator type is structurally incompatible; it fails because none of the recorded one- or two-accelerator systems reaches 20,000 output tokens/s. T6 fails for a different reason: the demand is above the fastest observed candidate even when eight accelerators are allocated. Reporting the failed conditions is more defensible than proposing an untested configuration.

Ranking must remain segregated from filtering. An option that fails a single strict condition cannot fix it by doing well on another. Not until all conditions clears does the standard rule lower accelerator count. The backup tie-break cuts out waste, which avoids showing unnecessary measured capacity as beneficial by default. Different organizations can switch the tie-break with power, budget, vendor policy, or risk once similar proof exists; the hard-constraint layer would remain exactly as it is.

6.2. Power and Cost Interpretation

S1 shows an critical evidence discipline. The 6.26-kW figure is taken from a comparable tracked test, rather than the chosen Dell, HPE, or Oracle MLPerf systems. Scaling it by nodes makes it easy to calculate an internally consistent stress test, but cannot reveal actual whole-system energy. Just like that, PUE converts IT power into a facility estimate; It isn't a real reading of the cooling plant. For electrical safety, vendor maximum or PSU-capacity evidence is the relevant conservative input. These quantities should never be collapsed into one column labelled simply "power."

Cost claims require the same validation. No numerical cloud comparison is included because the exact regional price record was not independently logged. Dell/HPE acquisition cost also depends on a complete configuration, support, region, taxes, and quotation date. In the absence of verified data, minimum-resource technically feasible is the correct label. Any claim that the framework identifies the most economical cloud or on-premise system would be baseless.

6.3. Relationship to the Prototype Catalogue

The catalogue documentation and experiments serve dif-ferent purposes. The catalogue is an implementation deliverables: after normalization and provenance review, it can hydrate filters for form factor, memory, electrical characteristics, networking, storage, racks, PDUs, and related equipment. S1 and S2 are research proof: they validate expected decisions for precisely defined workloads and can become validation gates for the future prototype.

decoupling the layers mitigates two errors. First, the exixtance of a product row does not mean its workload performance has been quantified. Second, a benchmark sub-mission does not provide every attribute needed for facility engineering. A production implementation should use concsiatent entity identifiers, normalize the vertically concatenated storage and networking tables, remove duplicate datasets, differentiate manufacturer facts from estimates,

and attach field-level lineage and retrieval dates. The current paper does not claim that this data-engineering work has been finalzed.

6.4. Threats to Validity and Limitations

The study has five primary constraints. First, it is a post-hoc, trace-driven assessment. ResNet-50 was not re-trained and Llama 2 70B inference was not run on local hardware. The study hence cannot empirically identify measurement error in the official results. Second, broader applicability is restricited by the chosen benchmark versions, models, datasets, precision/quality targets, systems, and software stacks. A different model or serving strategy may change the sequence.

Third, the candidate space is limited. The framework choses among recorded configurations; it does not forecast arbitary counts or merge components into an uevaluated system. This contraint reduces scope but preserves the credibility of the recommendation. Fourth, S1's energy values are early stage estimations and S2 has no operational-energy output. Comprehensive monitoring under synchronized conditions would be required for a comparative energy claim. Fifth, exact on-premise price optimization is unaddressed. Matched dated vendor bids are needed before a financial grading can be justified.

The experiment also uses benchmarking constants rather than requirements gathered from an operating organization. Deadlines, accelerator bounds, PUE, rack limits, and bud-gets therefore illustrate decision patterns, not standard scoping defaults. Regardless, the cases are reppricable, include extremes, and disclose rather than mask the assumptions on which the results lean.

6.5. Next Implementation and Research Steps

The upcoming operational task is to adapt the documented catalogue into a normalized, versioned schema and present the two simulation engines through one inter-face. The six S2 profiles, S1 deadline sweep, and regulated tests should be maintained as regression tests. A configuration should enter the performance catalogue only when its benchmark profile and provenance are finalized.

Extended research should add synchronized whole-system power metrics, replicable on-premise proposals, and extra workloads without changing theverification criteria. Only after those inputs are available should the framework streamline operational energy or total financial cost. A physical verification study could run smaller accessible models locally and cross reference recorded metrics with framework decisions. In-depth cooling, electrical backup, structural loading, and site design should remain subject to competent engineering scrutiny rather than being deduced from the current simulation.

7. Conclusion

This paper proposed WISP-DC, a workload-aware viability framework validated with official MLPerf traces and reproducible Python simulations. The training scenario demostrated that the minimum-accelerator recommendation for the same ResNet-50 workload moved from 4 to 8, 16, and 64 H100 GPUs as the deadline compresses; a two-minute target was

unattainable among the six profiled configurations. The inference scenario showed that Llama 2 70B demands must be analyzed collectively: 20,000 output tokens/s was in-feasible with at most two accelerators but feasible with eight, while 32,000 tokens/s was outside all four observed systems. The maintained data-backed realizations passed 24 automated audits in total, and S2 replicated all six predefined cases across an additional 99-case sensitivity grid.

The core finding is knowingly focused. Official benchmark proof can validate an verifiable early feasibility decision when each workload uses its own execution semantics, hard limits are applied before ranking, and omitted evidence is not replaced by an assumption. The paper verifies that decision process. It does not assert new model-performance measurements, full facility engineering, logged S2 energy, or the lowest cost on-premise purchase. Those boundaries make the current result useful as a defen-sible research baseline and a set of acceptance tests for the intended prototype.

Appendix A. Complete Validation Inventory

Tables 12 and 13 tabulate the performed audits maintained in the produced validation outputs. They are incorporated to make the meaning of "validated simulation" concrete. A pass indicates that the current code generated the slated targeted behavior; it does not mean that the test empirically repeated an MLPerf workload.

Table 12: S1 Controlled Validation Checks

ID	Test input or condition	Expected behavior	Result
CT-01	Two-minute training deadline	No evaluated configuration is fea-sible	Pass
CT-02	Three-minute training deadline	OCI-64G is the only feasible config-uration	Pass
CT-03	10-kW rack limit applied to 11.5-kW Dell planning maxi-mum	Candidate is rejected by rack-power con-straint	Pass
CT-04	Audit all raw Dell-8G statuses	Exactly one aborted log is exposed	Pass
CT-05	Increase PUE from 1.2 to 1.4 to 1.6	Facility-energy estimate increases strictly	Pass
CT-06	Request physical rack output for OCI	Outputremainsprovider-managed; no rack is fabricated	Pass

Appendix B. Worked Calculation and Decision Examples

For Dell-8G, the nominal duration is 13.16278 minutes, or 0.21938 hours. With the 6.26-kW mapped active substitute, the manual computation is

Table 13: S2 Automated Validation Checks

No.	Check	Purpose	Result
1	Four official configurations load	Detect missing or duplicated catalogue records	Pass
2	Throughput values equal retained offi-cial extracts	Detect transcription or parsing error	Pass
3	All baseline results are marked VALID	Enforce official validity status	Pass
4	Baseline records pass latency and qual-ity thresholds	Reconcile TTFT, TPOT, and ROUGE fields	Pass
5	T1–T6 match workbook expectations	Establish implementation-to-workbook agreement	Pass
6	T3 is infeasible at maximum two accel-erators	Confirm binding resource/throughput boundary	Pass
7	T4 becomes feasible when ceiling in-creases to eight	Confirm logical response to one changed input	Pass
8	Equal-count tie uses minimum excess throughput	Make ranking deterministic	Pass
9	H200-8 is selected when H100-8 nar-rowly misses	Test the 30,499–31,281 tokens/s interval	Pass
10	Requirement above all observations is infeasible	Prevent extrapolated capacity	Pass
11	Strict TTFT limit can reject every candi-date	Enforce TTFT independently	Pass
12	Strict TPOT limit can reject every candi-date	Enforce TPOT independently	Pass
13	Strict accuracy target can reject every candidate	Enforce quality independently	Pass
14	Manual tokens-per-accelerator arith-metic agrees	Check derived performance arithmetic	Pass
15	Manuallatency-marginarithmetic agrees	Check derived constraint margin	Pass
16	Monetary price optimization flag is false	Prevent unsupported cheapest-system claim	Pass
17	Non-positive throughput is rejected	Validate user input	Pass
18	Non-positive accelerator ceiling is re-jected	Validate user input	Pass

$$E_{\mathrm{IT}} = 6.26 \times 0.21938 = 1.37332 \text{ kWh}, \tag{10}$$

$$E_{\mathrm{facility}} = 1.4 \times 1.37332 = 1.92264 \text{ kWh}, \tag{11}$$

$$H = 6.26 \times 3412.14 = 21,360 \text{ BTU/h}. \tag{12}$$

These figures match the produced S1 output. The parity verifies the implementation of the arithmetic, not the a pplicability of the 6.26-kW proxy to the specific Dell record.

For S2 T4, GH200-2 fails to meet the targeted rate by $20,000 - 8,674.58 = 11,325.42$ output tokens/s and is dropped before ranking. H100-8 and H200-8 surpass the threshold by 10,498.87 and 11,281.09 output tokens/s, respectively. Be-cause both use eight accelerators, the secondary policy choses H100-8. If the demand increases to 31,000 output tokens/s, H100-8 misses by 501.13 while H200-8 holds a margin of 281.09; H200-8 is then the only viable observed option.

Appendix C. Prototype Catalogue Readiness

To protect the demarcation between prototype development and experimental testing, the prototype catalogue is considered independently from the S1 and S2 simulations. The catalogue provided for this study contained eleven CSV tables covering AI servers, GPUs, C-PUs, networking, storage, racks, PDUs, UPS equipment, cooling-related hardware, and power distribution. Two power-distribution tables contained redundant records; after eliminating that duplicate, ten unique datasets persisted. The AI-server table uses a broad product schema and includes entries for the Dell XE8640, Dell XE9680, and HPE Cray XD670, but it does not contain the DL384 Gen12 used in the GH200 inference cases. The GPU table includes the H100, H200, and GH200 product families.

The balance files require data engineering before they can securely generate recommendations. Networking and storage are vertically concatenated batches with different schemas rather than single flat entity tables. Some rack, UPS, cool-ing, and power fields are approximated, computed, conjected, or plainly "not stated." These are not flaws to mask; they define the next implementation directives. A production catalogue should (i) define one normalized schema per entity type, (ii) remove duplicate files and rows, (iii) add stable product identifiers, (iv) preserve units, region, currency, and effective date, (v) label manufacturer facts separately from estimates, and (vi) attach field-level provenance. Until that normalization is complete, the broader tables should sup-port documentation and manual audits rather than automated feasibility assertions.

The alignment between the two work streams is there-fore evident. The prototype catalogue feeds the future system's candidate and metadata layer. S1 and S2 supply validated decision rules and expected outputs. When the interface is implemented, the current scenarios become ac-ceptance tests: the prototype should reproduce the same decision whenever it receives the same versioned evidence and user constraints.

Data and Code Availability

The reproducibility package contains the S1 and S2 Python programs, the extracted official benchmark records with source URLs, declared scenario inputs, generated outputs, automated tests, figures, validation reports, and the recon-ciled workbook. Public benchmark records are available in the MLCommons Training v4.0 and Inference v5.0 reposi-tories [10], [11]. [REMAINING: code/data package in a versioned public repository URL .]

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